



A Microfluidic Device Enabling Single-Cell Isolation Using Hydrodynamic Trapping

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Abstract

Traditional single-cell isolation methods are often limited by high costs and complex automation, necessitating more accessible and scalable alternatives. To address this, we developed a passive microfluidic device utilizing optimized hydrodynamic trapping to isolate single particles in a 9 mm-spaced layout compatible with standard 96-well microtiter plates. Achieving up to 83% trapping efficiency for 40 μm PAM beads, this platform provides a low-cost solution for deterministic isolation without the need for active traps. Future work will expand the system to a full 96-well dispense platform, offering a scalable tool to improve access to high-throughput single-cell imaging and downstream analysis.

Background

Single-cell analysis is important for the study of individual cell behavior, whereas bulk measurements tends to mask meaningful cell-to-cell differences. Microfluidics enables precise control of small volumes and is a promising field for single cell isolation.

Current single-cell isolation methods rely on costly hardware, automation, and imaging. Other microfluidics methods for isolation are either not resettable, require low cell concentrations, non-deterministic, require active traps, or have no dispensing features.

Our project is to design and test a passive microfluidic device utilizing hydrodynamic trapping to isolate single cells aligned above a 96-well microtiter plate for downstream single-cell dispensing and imaging analysis. Our approach looks to provide a scalable and accessible isolation platform for high-throughput single-cell experiments.

Methods

Device Fabrication

Molds were fabricated on 4" silicon wafers using SU-8 2050 (~60 μm) via photolithography. Wafers were cleaned (acetone, IPA, DI), dehydrated (150 $^{\circ}\text{C}$, 10 min), spin-coated, soft-baked (65 $^{\circ}\text{C}$ 3 min; 95 $^{\circ}\text{C}$ 9 min), UV exposed (177.86 mJ/cm^2 , ~10 mW/cm^2 , 17.79 s), post-baked (65 $^{\circ}\text{C}$ 1.5 min; 95 $^{\circ}\text{C}$ 6 min), developed (10 min), and hard-baked (150 $^{\circ}\text{C}$, 10 min). PDMS (10:1) was cast, degassed, cured (65 $^{\circ}\text{C}$, >6 h), ports (0.75 mm) punched, and bonded to glass via oxygen plasma. Features were confirmed with an optical microscope and profilometer. PDMS (10:1 pre-polymer:curing agent) was cast, degassed, and cured (65 $^{\circ}\text{C}$, >6 h); ports (0.75 mm) punched, and bonded to glass via oxygen plasma.

Experimental Setup

40 μm Optiprep beads were prepared for testing. Devices were primed with DI water to remove air bubbles. A 1 mL syringe delivered beads through tubing to the inlet using a syringe pump starting at 200 $\mu\text{L}/\text{hr}$ and decreasing to 50 $\mu\text{L}/\text{hr}$ once beads entered the field of view. Flow behavior and hydrodynamic trapping/reset were monitored under an optical microscope.

Device

3 different trapping geometries reported in prior microfluidic studies were tested before selecting a candidate design. Main channel width was varied across Tan and Sun's trap designs (60, 80, 100, 120 μm), whereas Sochol's trap design was designed for specifically 85 μm beads. Each trap design was tested in a serial, back-to-back trapping format and a separate 9mm-spaced format, spacing that is consistent with the center-to-center spacing of wells in standard 96-well microtiter plates. The candidate trap design, optimized for isolation of 40 μm polyacrylamide (PAM) beads as cell analogs, spans half of the microtiter plate (48 wells).

Device operation relies on passive laminar flow and hydrodynamic resistance. Based on the "least flow resistance path" principle, flow initially favors the trap path such that incoming beads are directed into an empty trap. Once occupied, the trapped particle increases local resistance and redirects flow through the main channel and into the next available trap for sequential capture. Later iterations used rounded corners, adjusted trap position relative to the main channel, and reduced trap depth to better favor single-particle occupancy.

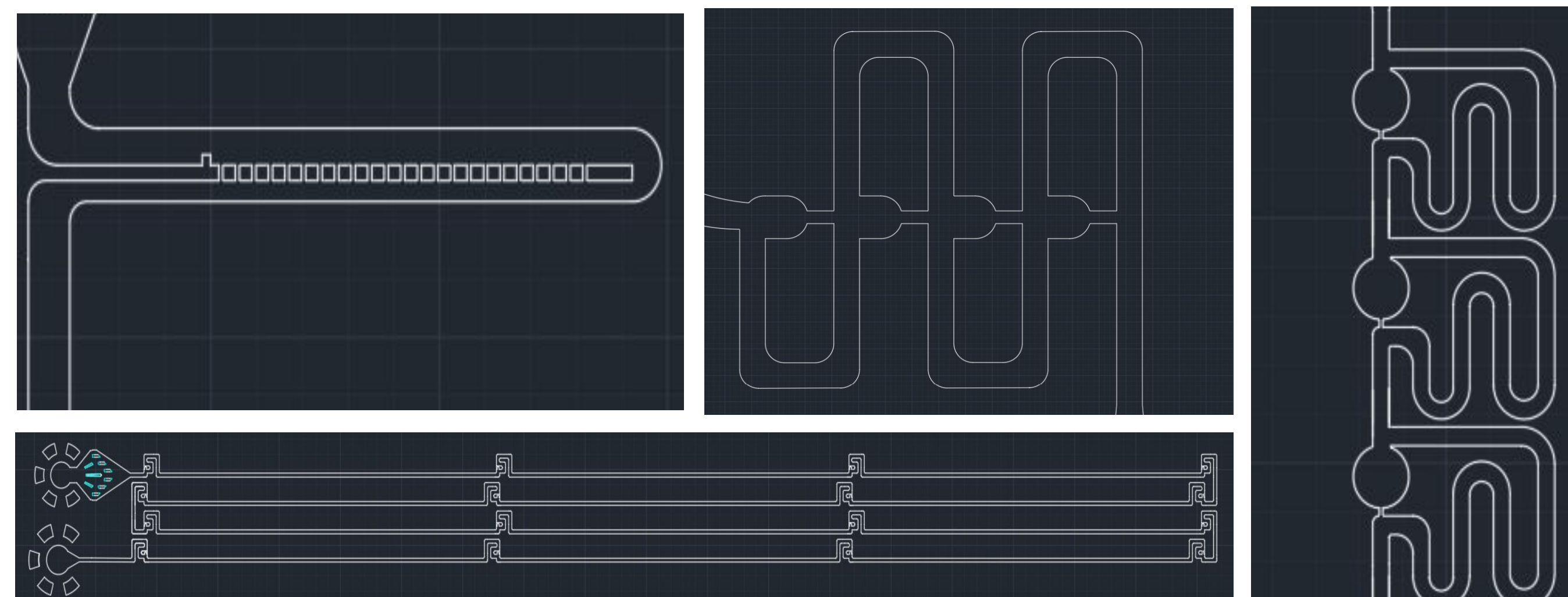


Fig 1. Serial trap designs inspired by Sochol et al. (top left), Tan et al. (top middle), Sun et al. (right)

Fig 2. 9mm-spaced trap design consistent with center-to-center spacing of 96-well microtiter plates Tan et al. (bottom)

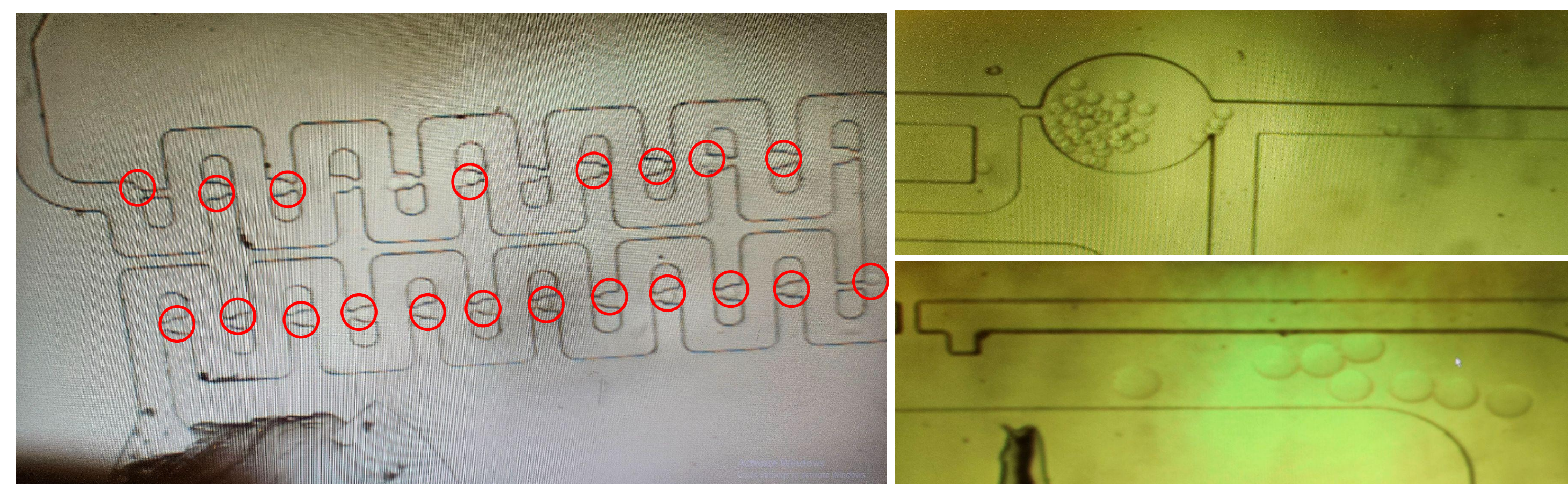


Fig 3. Microfluidic device with serial-traps. (left) Inspired by Tan et al., successful trapping events are circled in red tested on 40 μm PAM beads. (top right) Inspired by Sun et al., clustered trapping observed on 40 μm PAM beads. (bottom right) Inspired by Sochol et al., clogging observed in the main channel tested on 85 μm PAM beads.

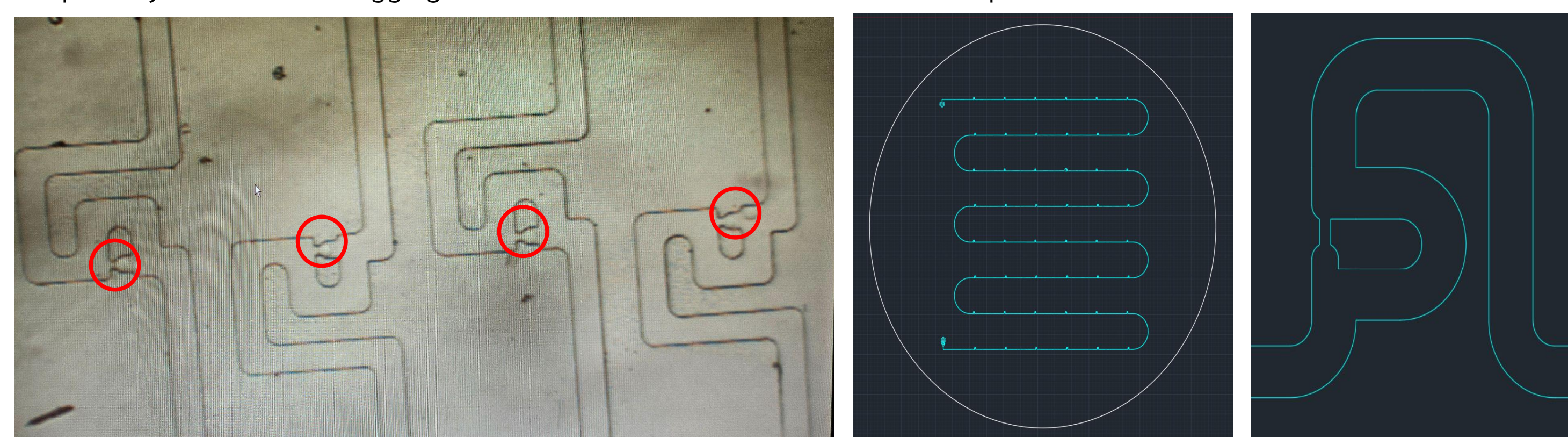


Fig 5. Microfluidic device with 9 mm-spaced layouts inspired by Tan et al. Successful bead trapping events are circled in red.

Fig 6. (left) Final mask design with 9 mm-spaced layouts with 48 trapping sites inspired by Tan et al. (right) Final Trapping Site inspired by Tan et al. optimized based on observations during testing.

Results & Discussion

Early prototypes with features <20 μm failed to resolve reliably, so later designs used features >20 μm . No successful trapping was observed in Sochol's and Sun's serial designs due to clogging and clustered trapping, respectively. In contrast, Tan's serial design achieved 83% single-bead trapping (20/24 traps), while the 9 mm layout achieved 81.25% (13/16) for 80 μm channels and 25% (4/16) for 120 μm channels; reset was successful using transient pressure pulses, with non-reversed flow outperforming reversed flow.

Bead isolation was achieved via passive hydrodynamic trapping using the least flow resistance principle. Oversized traps allowed multiple beads and fabrication limits prevented reliable <20 μm features. Rounded corners and centered traps improved capture. Device inconsistencies arose from spin coater variability, while air bubbles and debris caused inlet and channel clogging.

Future work includes developing a full 96-well trap-to-dispense platform requiring molds >4", and implementing a slip-lock PDMS lid for controlled dispensing. Additional efforts will focus on improving suspension quality, minimizing bubbles, testing with live cells, and optimizing PDMS stiffness and feature resolution.

Conclusion

This study successfully developed a passive microfluidic platform achieving up to 83% single-particle trapping efficiency through optimized hydrodynamic resistance and 9 mm-spaced geometries. By providing a deterministic isolation method without the need for complex automation, this device lays the foundation for a low-cost, scalable alternative to current high-end hardware. Future work to implement proper dispersions would allow this device to facilitate single-cell analysis and integration into standard downstream workflows. Ultimately, this platform has the potential to improve access to single-cell research by making high-throughput experimentation accessible to a wider range of laboratory settings.

References

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